RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

Started on	Thursday, 27 August 2026, 9:06 AM
State	Finished
Completed on	Thursday, 27 August 2026, 9:07 AM
Time taken	36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Recursive Binary Search function (Divide and Conquer approach)
int recursiveBinarySearch(int *arr, int low, int high, int target) {
    if (low > high) {
        return -1; // Element not found
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    } else if (arr[mid] > target) {
        return recursiveBinarySearch(arr, low, mid - 1, target);
    } else {
        return recursiveBinarySearch(arr, mid + 1, high, target);
    }
}

int main() {
    int n;
    // Read the size of the array
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));

    // Read n array elements (each on a new line as specified)
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int x;
    // Read the sum value x
    scanf("%d", &x);

    int found = 0;
    int elem1 = -1, elem2 = -1;

    // Iterate through each element and search for its complement using D&C
    for (int i = 0; i < n; i++) {
        int target = x - arr[i];
        // Search in the sub-array after the current element
        int j = recursiveBinarySearch(arr, i + 1, n - 1, target);

        if (j != -1) {
            elem1 = arr[i];
            elem2 = arr[j];
            found = 1;
            break;
        }
    }
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course